

19	Which of the following statements describing a gravitational and an electric field is correct?	
	A	The values of gravitational potential and electric potential are always negative.
	B	The direction of the gravitational field strength and electric field strength, at any point in their respective fields, always points towards decreasing potential values.
	C	The values of the gravitational potential energy and electric potential energy are always positive.

D	The direction of the gravitational field always points towards the point mass and the direction of the electric field always points towards the point charge.
	Answer: B Statement A is true for the gravitational potential but in the case of the electric potential, it depends on the charge that is within the electric field. Statement B is always true as the relationship between field strength and potential is that field strength is the negative of the potential gradient at that point. Statement C is true only for the gravitational potential energy but not the electric potential energy. Statement D is true for the gravitational field but not for the electric field as it depends on the polarity of the charge.

20	A potential difference V is applied between two parallel plates a small distance d apart, and
----	---